

MarineGuard AI — Investigation Report

SYNTHETIC DEMONSTRATION DATA — not real-world observations.

1. Incident information

Field	Value
Incident ID	MG-LIVE-20260906
Title	Arabian Sea Active Operational Incident (Live Feeds)
Location	Arabian Sea, Mumbai Offshore Corridor
Observation time	2026-09-06T19:33:23Z
Provenance	SYNTHETIC DEMONSTRATION DATA

2. Satellite observation

SAR scene associated with this incident. Detection mode: **U-Net ML Inference**. Trained U-Net weights were used for segmentation.

3. Spill detection

Field	Value
Mode	U-Net ML Inference
Confidence	75.8%
Status	Completed

4. Spill geometry

Field	Value
Area	88.71 km²
Length	12.47 km
Width	9.62 km
Perimeter	39.66 km
Centroid	19.07793, 71.82779
Shape	compact

5. Probable origin

Field	Value
Latitude	19.1922
Longitude	71.75352
Estimated origin time	2026-09-06T15:03:23Z
Window	2026-09-06T13:33:23Z → 2026-09-06T16:33:23Z
Uncertainty radius	7.8 km

Probable origin is an analytical backward integration combining surface current vectors and 3% wind leeway with Coriolis deflection. It defines a probable search window, not an exact spill location.

6. Hindcast result

Backward trajectory points: 25. Model: backward_euler_drift_integration.

7. Forecast

Lead time	Latitude	Longitude
+0 h	19.07793	71.82779
+6 h	19.13083	72.07512
+12 h	18.82211	72.14498
+24 h	18.4317	72.21457
+48 h	17.3946	72.06642

Forward trajectory forecast incorporates predicted hydrodynamic surface current and 3% wind leeway. Uncertainty corridor expands over time due to oceanic turbulence and atmospheric variation.

8. AIS traffic analysis

Field	Value
Total vessels	5
Spatial candidates	5
Temporal candidates	5
Final candidates	5

9. Candidate vessel ranking

Rank	MMSI	Score	Priority
1	419000222	77.1	HIGH
2	419000111	70.9	MEDIUM
3	419000444	65.5	MEDIUM
4	419000555	63.0	MEDIUM
5	419000333	54.8	MEDIUM

10. Evidence

MV COASTAL PEARL (419000222) — Potential Suspect Vessel (Investigation Priority 1)

- Closest approach: 1.8 km from probable origin region.
- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 30.7° relative to inferred drift axis.
- Operating speed: min 12.0 kn, mean 12.0 kn.
- Vessel type relevance: cargo (operational context score 55%).

MT SYNTHETIC STAR (419000111) — Evidence-Supported Candidate #2

- Closest approach: 13.1 km from probable origin region.
- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 48.2° relative to inferred drift axis.
- Operating speed: min 8.4 kn, mean 8.4 kn.
- Vessel type relevance: tanker (operational context score 100%).

MV NORTHBOUND (419000444) — Evidence-Supported Candidate #3

- Closest approach: 3.8 km from probable origin region.

- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 71.6° relative to inferred drift axis.
- Operating speed: min 14.5 kn, mean 14.5 kn.
- Vessel type relevance: cargo (operational context score 55%).

MT ARABIAN SHUTTLE (419000555) — Evidence-Supported Candidate #4

- Closest approach: 11.7 km from probable origin region.
- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 79.5° relative to inferred drift axis.
- Operating speed: min 9.2 kn, mean 9.2 kn.
- Vessel type relevance: tanker (operational context score 100%).

FV WESTERN NET (419000333) — Evidence-Supported Candidate #5

- Closest approach: 24.6 km from probable origin region.
- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 35.4° relative to inferred drift axis.
- Operating speed: min 6.5 kn, mean 6.5 kn.
- Vessel type relevance: fishing (operational context score 30%).

11. Confidence

Detection confidence 75.8%. Drift uncertainty radius 7.8 km. Forecast corridor width 23.6 km.

12. Data gaps

- Satellite detection may include false positives, especially in prototype/demo mode.
- AIS coverage may be gapped, delayed, or spoofed.
- Ocean wind/current fields are sparse demonstration vectors, not a circulation model.

13. Responsible-AI disclaimer

This system prioritises vessels for investigation and does not establish definitive legal responsibility. The highest-ranked vessel is a highest-priority candidate only. Results require human validation and must not be used as the sole basis for legal enforcement.